Customer Shopping Behavior Analysis

Project Report | Python • MySQL • Power BI

1. Project Overview

This project analyzes customer shopping behavior using a dataset of 3,900 customer records (one purchase record per unique customer) across four product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions on marketing, pricing, discounts and product strategy.

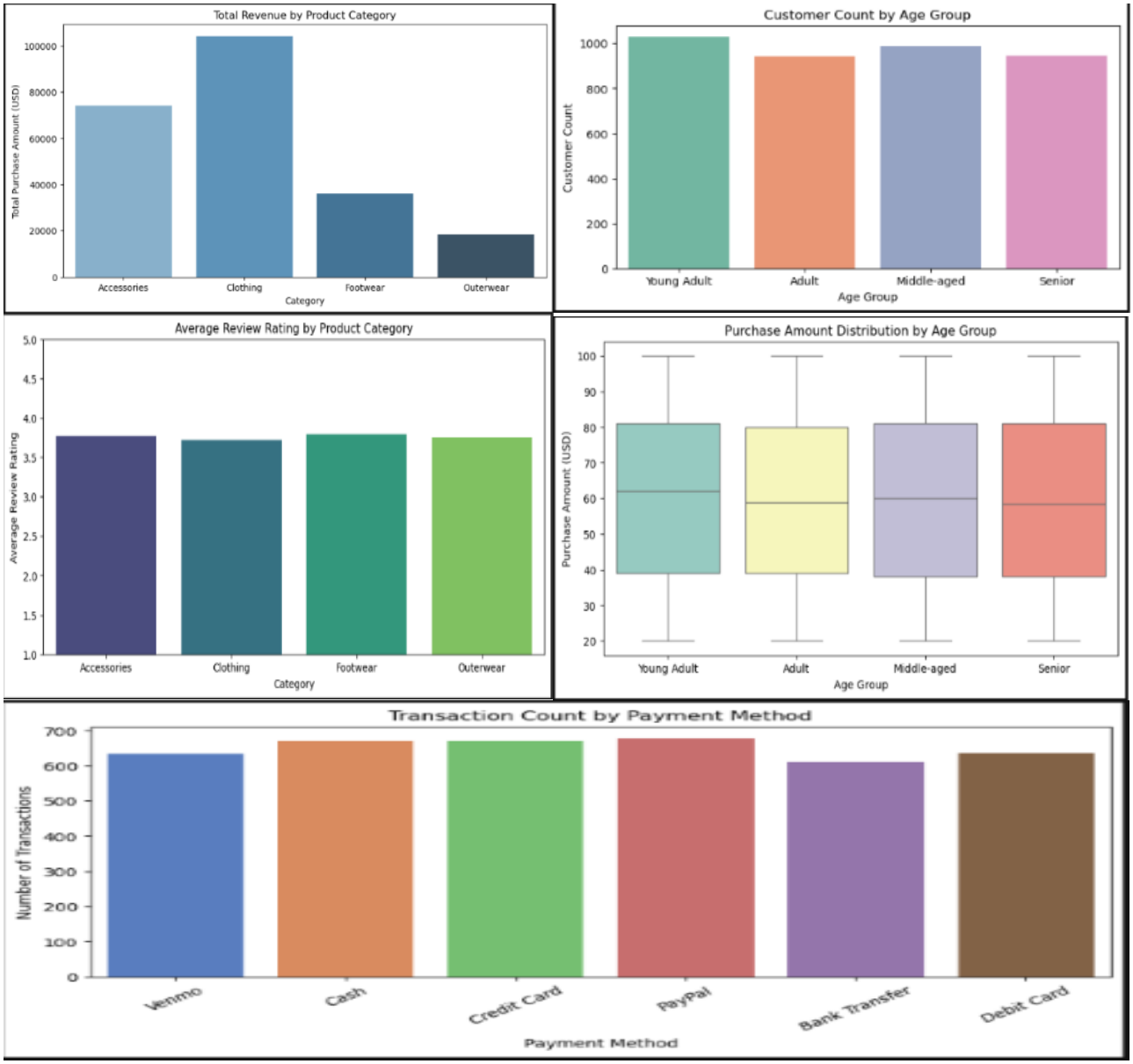
2. Dataset Summary

- **Rows:** 3,900 (3,900 unique Customer IDs) **Columns:** 18 in the raw file (19 after cleaning and feature engineering)
- **Key features:**
  - Customer demographics: Age (18 to 70), Gender, Location (50 U.S. states), Subscription Status
  - Purchase details: Item Purchased (25 items), Category (Clothing, Accessories, Footwear, Outerwear), Purchase Amount (USD), Season, Size, Color
  - Shopping behavior: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type (6 options), Payment Method (6 options)
- **Missing data:** 37 values in the Review Rating column (no other column has missing values)

3. Exploratory Data Analysis using Python

Data preparation and cleaning were carried out in Python (pandas):

- **Data loading:** imported the dataset using pandas.
- **Initial exploration:** used `df.info()` to check structure and `df.describe()` for summary statistics. Purchase amounts range from $20 to $100 (mean $59.76); customer ages range from 18 to 70 (mean 44.1).
- **Missing data handling:** the 37 missing Review Rating values were imputed with the median rating of each product category.
- **Column standardization:** renamed columns to snake_case (for example `Purchase Amount (USD)` became `purchase_amount`).
- **Feature engineering:**
  - Created **age_group** by splitting ages into four quartile-based bins: Young Adult (18-31), Adult (32-44), Middle-aged (45-57) and Senior (58-70).
  - Created **purchase_frequency_days** by mapping Frequency of Purchases to days (Weekly = 7, Fortnightly and Bi-Weekly = 14, Monthly = 30, Quarterly and Every 3 Months = 90, Annually = 365).
- **Data consistency check:** verified that `discount_applied` and `promo_code_used` are identical in all 3,900 rows, so the redundant `promo_code_used` column was dropped.
- **Visual exploration:** plotted revenue by category, customer distribution by age group, ratings by category, purchase amounts by age group and transactions by payment method (below).
- **Database integration:** connected the Python script to MySQL (SQLAlchemy and PyMySQL) and loaded the cleaned DataFrame into the `customer` table of the `customer_behavior` database for SQL analysis.



A Figure 1. Exploratory plots from the Python notebook (cleaned dataset).

4. Data Analysis using SQL (Business Transactions)

Structured analysis was performed in MySQL on the cleaned `customer` table to answer 20 key business questions. Overall totals for reference: **$233,081** total revenue, **$59.76** average purchase amount, **3.75** average review rating.

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.

Gender	Customers	Revenue	Avg spend
Female	1,248	$75,191	$60.25
Male	2,652	$157,890	$59.54

Insight: Male customers generate 67.7% of revenue, but they are also 68.0% of customers. Average spend per customer is almost the same (Female $60.25, Male $59.54), so the gap comes from customer volume, not spending level.

2. High-Spending Discount Users – Identified customers who used a discount but still spent at or above the average purchase amount ($59.76).

customer_id	purchase_amount
2	$64
3	$73
4	$90
7	$85
9	$97

Insight: 839 customers (first 5 shown) fit this profile, which is 50.0% of the 1,677 purchases made with a discount. Discounts do not stop half of these buyers from spending above average.

3. Top 5 Products by Rating – Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Insight: Ratings are tightly clustered: the top five span only 0.08 points. Gloves, Sandals and Boots lead.

4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

shipping_type	Average purchase amount
Standard	$58.46
Express	$60.48

Insight: Express buyers spend $2.02 more on average than Standard buyers. Across all six shipping types the averages range from $58.46 (Standard) to $60.73 (2-Day Shipping), a narrow spread.

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.

subscription_status	Customers	Avg spend	Total revenue
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Insight: Subscribers are 27.0% of customers and bring 26.9% of revenue. Their average spend is not higher than non-subscribers' ($59.49 vs. $59.87).

6. Discount-Dependent Products – Identified the 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate (%)
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

Insight: Overall, 43.0% of all purchases (1,677 of 3,900) use a discount. These five products are the most discount-dependent, with roughly half of their sales discounted.

7. Customer Segmentation – Classified customers into New (1 previous purchase), Returning (2 to 10) and Loyal (more than 10) segments.

customer_segment	Number of Customers	Share
Loyal	3,116	79.9%
New	83	2.1%
Returning	701	18.0%

Insight: About four in five customers already fall in the Loyal segment, and only 83 customers are New. With these thresholds, growth will come mainly from acquiring new customers and moving Returning customers up.

8. Top 3 Products per Category – Listed the most purchased products within each category.

Rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Pants	171
2	Clothing	Blouse	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Insight: Outerwear contains only two items (Jacket and Coat). Ties exist (Pants and Blouse both have 171 orders; Sunglasses and Belt both have 161), so their rank order is arbitrary in the query output.

9. Repeat Buyers & Subscriptions – Checked whether customers with more than 5 previous purchases are more likely to subscribe.

subscription_status	repeat_buyers
No	2,518
Yes	958

Insight: Of 3,476 repeat buyers, 27.6% subscribe, compared with 27.0% across all customers. Repeat buyers are only marginally more likely to subscribe, so subscription is not yet strongly tied to loyalty.

10. Revenue by Age Group – Calculated the total revenue contribution of each age group.

age_group	Total revenue	Share	Customers
Young Adult (18-31)	$62,143	26.7%	1,028
Middle-aged (45-57)	$59,197	25.4%	986
Adult (32-44)	$55,978	24.0%	942
Senior (58-70)	$55,763	23.9%	944

Insight: Young Adults contribute the most revenue and also have the most customers. The gap between age groups is modest (23.9% to 26.7% of revenue).

11. Payment Methods by Revenue – Ranked payment methods by total revenue.

payment_method	Transactions	Total revenue
Credit Card	671	$40,310
PayPal	677	$40,109
Cash	670	$40,002
Debit Card	636	$38,742
Venmo	634	$37,374
Bank Transfer	612	$36,544

Insight: Credit Card leads at 17.3% of revenue, but all six methods are close (Bank Transfer is lowest at 15.7%). No single payment method dominates.

12. Revenue by Season – Identified which season generates the highest total revenue and average purchase amount.

season	Orders	Total revenue	Avg purchase
Fall	975	$60,018	$61.56
Spring	999	$58,679	$58.74
Winter	971	$58,607	$60.36
Summer	955	$55,777	$58.41

Insight: Fall is the strongest season on both total revenue and average purchase amount. Summer is the weakest on both.

13. Revenue and Rating by Size – Calculated total revenue and average rating for each product size.

size	Items sold	Total revenue	Avg rating
M	1,755	$105,167	3.72
L	1,053	$61,667	3.75
S	663	$40,468	3.80
XL	429	$25,779	3.80

Insight: Size M accounts for 45.1% of revenue. Smaller sales volumes (S and XL) come with slightly higher average ratings.

14. Top 5 Locations by Revenue – Identified the five states generating the highest total revenue.

location	Customers	Total revenue
Montana	96	$5,784
Illinois	92	$5,617
California	95	$5,605
Idaho	93	$5,587
Nevada	87	$5,514

Insight: The top five states together make up only 12.1% of revenue, so sales are spread widely across all 50 states rather than concentrated in a few.

15. Ratings by Subscription Status – Compared review ratings between subscribed and non-subscribed customers.

subscription_status	Avg review rating	Total reviews
No	3.75	2,847
Yes	3.75	1,053

Insight: Satisfaction is identical for both groups (3.75), so subscription status does not change how customers rate their purchases.

16. Average Purchase Amount by Category – Ranked product categories by average purchase amount.

category	Items	Avg purchase	Total revenue
Footwear	599	$60.26	$36,093
Clothing	1,737	$60.03	$104,264
Accessories	1,240	$59.84	$74,200
Outerwear	324	$57.17	$18,524

Insight: Footwear has the highest average ticket, while Clothing earns the most in total (44.7% of revenue). Outerwear has both the lowest average ($57.17) and the smallest revenue share (7.9%).

17. Purchase Frequency and Spending – Analyzed how reported purchase frequency (in days) relates to total spending.

purchase_frequency_days	Customers	Total revenue	Avg spend
90	1,147	$68,859	$60.03
14	1,089	$65,207	$59.88
365	572	$34,419	$60.17
30	553	$32,810	$59.33
7	539	$31,786	$58.97

Insight: The 90-day and 14-day groups lead on revenue mainly because they contain the most customers (each group combines two frequency labels). Average spend per customer is nearly flat ($58.97 to $60.17).

18. Above-Average Spenders with 5-Star Reviews – Found customers who spent above the overall average and also left a 5-star review.

customer_id	item_purchased	purchase_amount	review_rating
3244	Hoodie	$99	5.0
1497	T-shirt	$98	5.0
1278	Blouse	$97	5.0
2174	Gloves	$97	5.0
2394	Socks	$97	5.0

Insight: 37 customers meet both conditions (top 5 by amount shown). They are a natural pool for reviews, referrals and loyalty outreach.

19. Most Popular Color per Category – Identified the most popular item color in each product category.

category	color	color_count
Accessories	Olive	66
Clothing	Teal	84
Footwear	Olive	31
Outerwear	Olive / Blue (tied)	19

Insight: Olive is the top color in three of four categories. In Outerwear, Olive and Blue tie at 19 each, so the query returns just one of them.

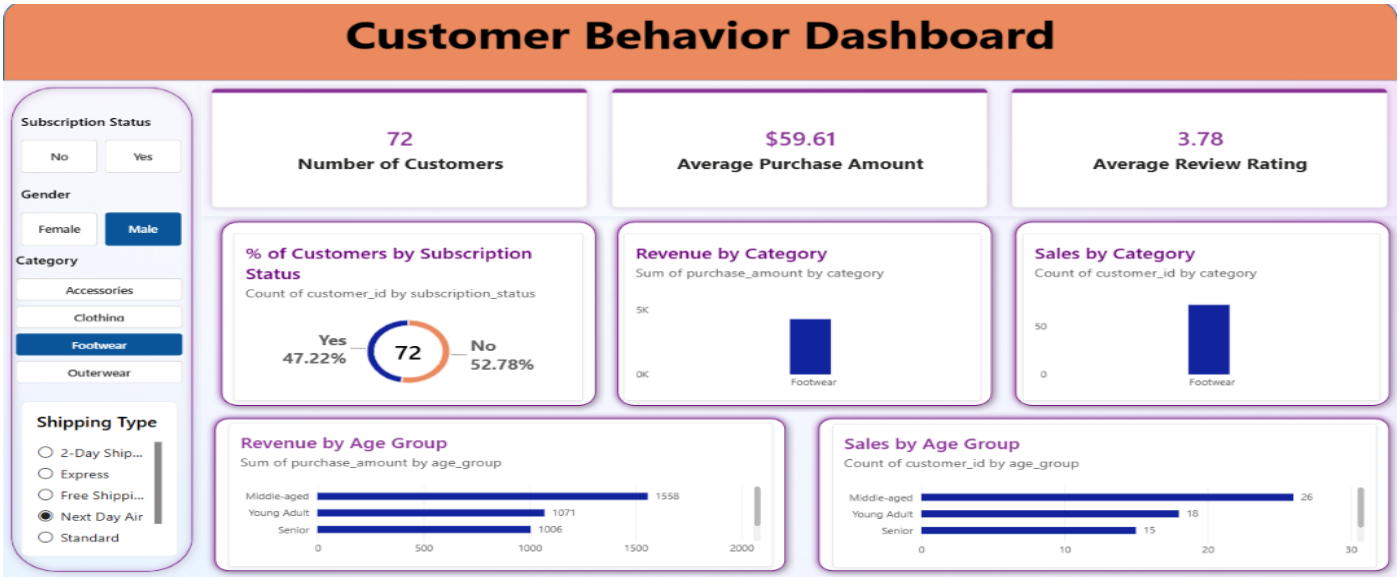
20. Categories Above the Overall Average – Found categories whose average purchase amount is higher than the overall average ($59.76).

category	category_avg_spend
Footwear	$60.26
Clothing	$60.03
Accessories	$59.84

Insight: Three categories sit above the overall average, though only by cents. Outerwear ($57.17) is the one below it.

5. Dashboard in Power BI

Finally, an interactive dashboard was built in **Power BI** to present these insights visually for stakeholders.



Headline figures from the cleaned dataset that the dashboard summarizes:

Metric	Value	Metric	Value
Number of customers	3,900	Subscribed customers	27.0%
Total revenue	$233,081	Purchases with a discount	43.0%
Average purchase amount	$59.76	Average review rating	3.75

6. Business Recommendations

1. Make subscriptions worth more. Subscribers (27.0% of customers) spend no more than non-subscribers ($59.49 vs. $59.87), rate products the same (3.75) and are barely more common among repeat buyers (27.6% vs. 27.0%). Design benefits that raise basket size, such as free shipping above a spend threshold or early access to new items, and A/B test them.

2. Build the loyalty program around the Returning segment. 79.9% of customers are already Loyal and only 2.1% are New. Reward the 701 Returning customers to move them up, and invest in acquiring new customers. Consider percentile-based thresholds so the program can distinguish a true top tier.

3. Review the discount policy. 43.0% of purchases carry a discount, up to about half of sales for Hat, Sneakers, Coat, Sweater and Pants. Discounted purchases average $59.28 vs. $60.13 without a discount, so discounts are not lifting order value. Test reducing discounts on these products to protect margin.

4. Position products using ratings and sales data. Feature top-rated items (Gloves, Sandals, Boots) and best-sellers (Jewelry, Pants, Blouse) in campaigns. Outerwear is the weakest category (7.9% of revenue, lowest average ticket), so consider bundles with Clothing to lift it.

5. Time campaigns to Fall. Fall leads on both revenue ($60,018) and average purchase ($61.56), while Summer is lowest. Plan stock and promotional spend ahead of Fall, and use Summer for targeted offers to lift the weakest season.

6. Target marketing with realistic expectations. Young Adults (18-31) contribute the most revenue (26.7%), and women are only 32% of customers despite a slightly higher average spend ($60.25 vs. $59.54), which points to an acquisition opportunity. Express and 2-Day shipping users spend slightly more than Standard users ($60.48 and $60.73 vs. $58.46), but the gaps are small.

Note on interpretation: most differences between groups in this dataset are small (typically under 5% on average spend), so the recommendations above should be validated with controlled tests before large investments. Figures were computed from the cleaned dataset after imputing 37 missing review ratings.